Math 142, S14 **Name:** Answer

Quiz 4 **Section:** **SSN:** xxx-xx-

<u>Instructions:</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (6 points) Evaluate $\int \frac{x-9}{(x+5)(x-2)}\,dx$ using the partial fraction decomposition.

$$\frac{x-9}{(x+5)(x-2)} = \frac{A}{x+5} + \frac{B}{x-2} = \frac{A(x-2)+B(x+5)}{(x+5)(x-2)} = \frac{(Ax-2A)+(Bx+5B)}{(x+5)(x-2)}$$

$$= \frac{(A+B)x+(-2A+5B)}{(x+5)(x-2)}$$

$$\Rightarrow \begin{cases} A+B=1 \\ -2A+5B=-9 \end{cases} \Rightarrow A=2,\ B=-1$$

Thus

$$\int \frac{x-9}{(x+5)(x-2)}\,dx = \int \left(\frac{2}{x+5} - \frac{1}{x-2}\right)dx$$

$$= 2\ln|x+5| - \ln|x-2| + C$$

Problem 2. (4 points) Evaluate the integral $\int e^{\sqrt{x}}\,dx$. (Hint: *First use the substitution* $u = \sqrt{x}$.)

Let $y=\sqrt{x}$, then $x=y^2$, $dx = 2y\,dy$. Thus

$$\int e^{\sqrt{x}}dx \overset{y=\sqrt{x}}{=\!=} \int e^{y}\,2y\,dy$$

$$= 2\int ye^{y}dy$$

$$= 2\left[ye^{y} - \int e^{y}dy\right] + C$$

$$= 2(ye^{y} - e^{y}) + C$$

$$= 2e^{y}(y-1) + C$$

$$= 2e^{\sqrt{x}}(\sqrt{x}-1) + C.$$